THE UNIVERSITY OF HONG KONG
FACULTY OF BUSINESS AND ECONOMICS

Master of Business Administration
PMBA6076 – AI and Investment Banking: M&A, Valuation, and LBOs
Academic Year 2026-2027

I. Information on Instructor and Tutor

Instructor: Prof. Matthias BUEHLMAIER
Email: buehl-teaching[at]hku.hk
Office: Room 1106, KKL Building
Phone: +852 2219 4177
Consultation times: Wednesdays from 4:00 p.m. to 7:00 p.m. To allow for better preparation, students should email the instructor a brief description of the consultation topics they wish to discuss at the latest on the day before the consultation. Students who cannot attend the consultation hours due to a time clash with other courses/tutorials may request a separate consultation time.

Tutor: Mr. Clive HO Man Chung
Office: Room 1026, KKL Building
Phone: +852 2857 8514
E-Mail: hoclive[at]hku.hk
Office hour: TBA

Pre-requisites: None

Recommended textbook: "Investment Banking: Valuation, LBOs, M&A, and IPOs" by Joshua Rosenbaum and Joshua Pearl (Wiley, university third edition, 2020). If available, please obtain the *university* edition since it contains additional materials such as end-of-chapter questions. (Copies of end-of-chapter questions will also be provided on the Moodle course website.)

II. Course Description and Objectives

Description:
This course is designed to develop a solid understanding of commonly discussed and applied issues in mergers and acquisitions (M&As). The topics covered in this course include the M&A process, methods of valuing a target firm, valuing synergies, the form of payment and financing, assessing highly levered transactions, governance in M&A, and M&A negotiations. M&A cases will be used in discussing these topics. By going through analyses and discussions of real-life M&As, students will gain experience in the application of financial theories and techniques to evaluate M&A decisions and transactions. The course integrates AI tools throughout the deal process, including LLM-assisted comparable company screening and AI-supported due diligence on company filings. The focus remains on M&A and valuation; AI is used as a practical tool, and no programming background is required.

Objectives:
1. Cultivate a deep and rich understanding of the M&A process from inception to completion.
2. Foster awareness of potential pitfalls in M&A.
3. Develop analytic competence to empower students to advance their careers in M&A, investment banking, and related fields.
4. Use AI tools critically and effectively in deal analysis.

III. Course Learning Outcomes (CLOs)

CLO 1	Understand the fundamental concepts and processes in M&A.
CLO 2	Value a target company using several valuation methods.
CLO 3	Discern the conceptual and technical differences between strategic mergers and leveraged

	buyouts.
CLO 4	Construct detailed financial models, such as merger consequences analysis, using both traditional tools and AI assistance.
CLO 5	Convincingly communicate and present one's evaluation and analysis of a proposed merger deal.

IV. Alignment of Programme and Course Learning Outcomes

MBA Programme Learning Outcomes	Aligned Course Learning Outcomes (CLOs)
PLO 1. Acquisition and internalization of business knowledge and skills in key functional areas	CLOs 1-4
PLO 2. Application of business knowledge to solve business problems	CLO 5
PLO 3. Development of managerial leadership and inculcating professional ethics and competency in business	CLOs 1-5
PLO 4. Development of global outlook	CLOs 1-5
PLO 5. Mastering communication skills	CLO 5

V. Teaching and Learning Activities (TLAs)

TLA#	Content	Aligned CLOs
TLA 1	Lectures: The instructor will hold lectures on major concepts and issues.	CLOs 1-4
TLA 2	Cases: Students will be asked to solve cases by groups and submit written reports. The objective of these cases is to give students an opportunity to study M&A in depth and then apply concepts and tools on real-world problems.	CLOs 4-5
TLA 3	Homework assignments	CLOs 1-4
TLA 4	Tutorials and consultation: The teaching assistant will hold tutorials to address students' questions and illustrate important examples to highlight major points covered in class.	CLOs 1-4
TLA 5	Class participation: Students are encouraged to raise questions in class and actively participate in the class discussion.	CLO 5
TLA 6	Group projects and in-class presentations, including an AI-supported deal analysis component.	CLOs 4-5

VI. Assessment

A#	Content	Weight	Aligned CLOs
A1	Written case report	15%	CLOs 4-5
A2	Midterm	35%	CLOs 1-4
A3	Group project presentation (individually evaluated)	15%	CLOs 4-5
A4	Group project report (evaluated on group level)	35%	CLOs 4-5

VII. Standard for Assessment

- **Course Grade Descriptors**

A	Excellent performance in all areas of the course.
B	Good performance in most areas of the course.
C	Satisfactory performance in some areas of the course.
D	Poor performance in most areas of the course.
F	Unsatisfactory performance in most areas of the course.

- **Assessment Rubrics for Each Assessment**

Grade	Course Grade Descriptor
Lecture Attendance, Participation, and Cases (a separate sheet with detailed requirements for the cases will be distributed)	
A	Well prepared for class discussion, very active in sharing views.
B	Reasonably prepared for class discussion, quite active in sharing views.
C	Not well prepared for class discussion, limited sharing of views.
D	Not well prepared for class discussion, no sharing of views.
F	No preparation for class discussion, no sharing of views.
Assignments and Exam(s)	
A	Excellent or very good replies to all criteria.
B	Good to very good replies to some or all criteria.
C	Fair to good replies to some or all criteria.
D	Poor replies to some or all criteria.
F	Very poor replies to some or all criteria, or failure to submit the assignment(s)/exam(s)
Group Project (a separate sheet with detailed requirements will be distributed)	
A	Excellent or very good ratings on all criteria.
B	Good to very good ratings on some or all criteria.
C	Fair to good ratings on some or all criteria.
D	Fair ratings on all criteria.
F	Absent from presentation or fail to submit the group report.

VIII. Means/Processes for Student Feedback on Course

- Class discussion feedback.
- A survey will be distributed to students around the date of the midterm to provide additional feedback on the course and teaching.

IX. Academic Integrity

The University Regulations on academic dishonesty will be strictly enforced. Please check the University Statement on plagiarism on http://www.hku.hk/plagiarism/.

Cheating or plagiarism of any kind would result in an automatic F grade for the course plus strict enforcement of all Faculty and/or University regulations regarding such behavior. Incident(s) of academic dishonesty will NOT be tolerated.

Academic dishonesty is a behavior in which a deliberately fraudulent misrepresentation is employed in an attempt to gain undeserved intellectual credit, either for oneself or for another. It includes, but is not necessarily limited to, the following types of cases:

a. Plagiarism - The representation of someone else's ideas as if they are one's own. Where the arguments, data, designs, etc., of someone else are being used in a paper, report, oral presentation, or similar academic project, this fact must be made explicitly clear by citing the appropriate

references. The references must fully indicate the extent to which any parts of the project are not one's own work. Paraphrasing of someone else's ideas is still using someone else's ideas, and must be acknowledged.

b. Unauthorized Collaboration on Out-of-Class Projects - The representation of work as solely one's own when in fact it is the result of a joint effort.

c. Cheating on In-Class Exams - The covert gathering of information from other students, the use of unauthorized notes, unauthorized aids, etc.

d. Unauthorized Advance Access to an Exam - The representation of materials prepared at leisure, as a result of unauthorized advance access (however obtained), as if it were prepared under the rigors of the exam setting. This misrepresentation is dishonest in itself even if there are not compounding factors, such as unauthorized uses of books or notes.

X. Course Schedule

The schedule is tentative and subject to change. For further details on e.g. case studies, midterm, and presentations, please see the relevant handouts.

Lecture	Date	Topic
1	Jan 16 a.m.	Introduction; AI in the deal process; sell-side M&A
2	Jan 16 p.m.	Sell-side M&A; trading comps; AI-assisted comps screening
3	Jan 23 a.m.	Trading comps; transaction comps; DCF
4	Jan 23 p.m.	**Case discussion (Pinkerton (A))**; DCF; AI-assisted due diligence
5	Jan 30 a.m.	**Midterm**; DCF; LBOs
6	Jan 30 p.m.	LBOs; LBO analysis
7	Feb 20 a.m.	LBO analysis; buy-side M&A
8	Feb 20 p.m.	Buy-side M&A; **group presentations**

XI. Additional Course Information